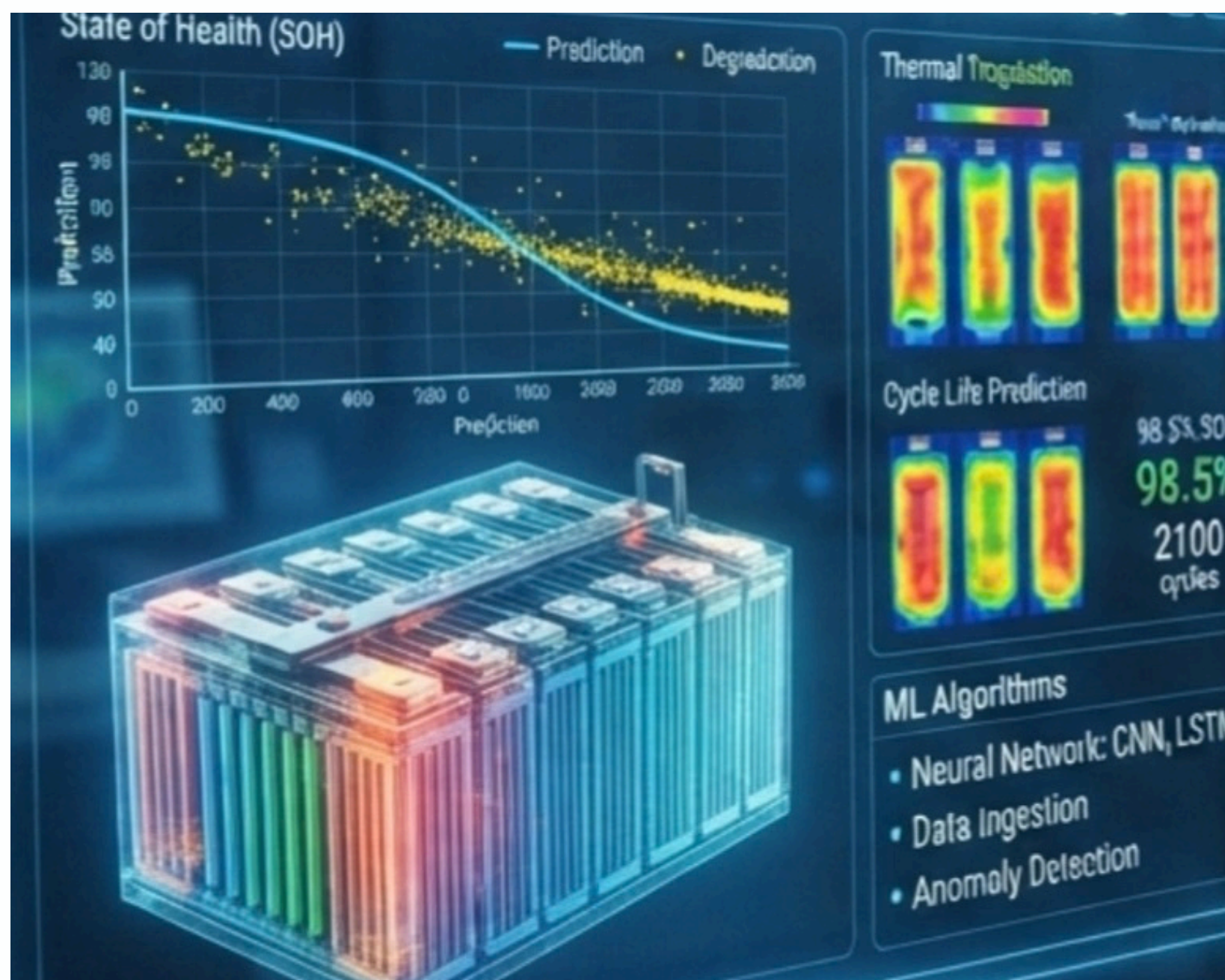




CrystalVolt

Machine Learning the Voltage of Electrode Materials in Metal-ion Batteries

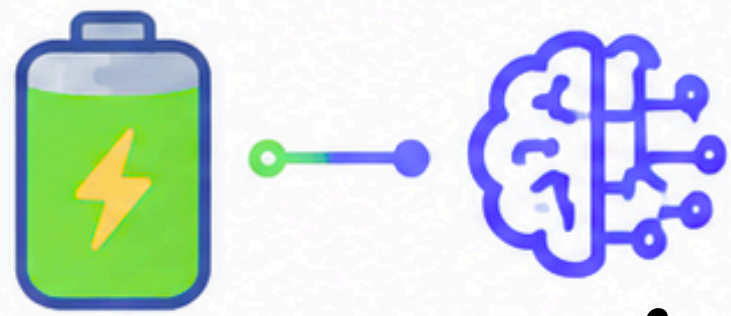


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Understanding Of Base Paper

Key Insights:

- The paper uses Machine Learning to predict the voltage of electrode materials.
- Material information such as chemical and elemental properties is converted into numerical features.
- ML models learn the relationship between these features and electrode voltage.

The Challenge:

- Finding the voltage of thousands of potential electrode materials is time-consuming and computationally expensive.
- The paper addresses this by using Machine Learning for fast voltage prediction.
- The dataset has an uneven distribution of metal-ion types, with some metals having significantly fewer data samples than others, which may affect the model's prediction performance for underrepresented metals.



Problem Definition

To develop a machine learning-based approach that can quickly and accurately predict the voltage of electrode materials for metal-ion batteries, reducing the time and computational cost of material screening.

Why it Is Important & Relevant:

- Saves time and cost by reducing repeated experiments.
- Predicts battery voltage using material features and ML.
- Helps researchers find promising materials faster.
- Supports the development of better batteries.



In-Scope

- Material & crystal property representation
- Feature engineering for ML
- Average electrode voltage prediction
- ML model comparison and evaluation
- Prediction for unseen electrode materials
- Voltage-based material screening

Out-Of-Scope

- Experimental material testing
- New DFT calculations
- Battery manufacturing
- Cycle-life, safety and cost prediction
- Experimental validation of predicted materials



Proposed Solution

Our Approach:

Material Data Preprocessing

- Clean missing and duplicate data
- Standardize material information
- Prepare data for ML models

Feature Engineering

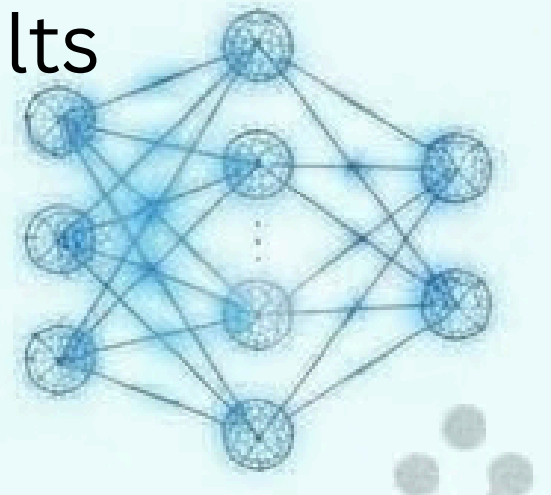
- Extract useful material features
- Convert features into numerical form
- Select important features for prediction

Machine Learning Models

- Random Forest → Robust prediction
- XGBoost → High accuracy
- Linear Regression → Baseline model

Voltage prediction

- Predict average electrode voltage
- Use material features as input
- Display predicted voltage in volts (V)



Expected Outcome:

Accurate prediction of electrode voltage

- Predict electrode voltage accurately
- Compare predicted and actual voltage
- Evaluate model accuracy

Fast Prediction for new materials

- Enter new material details
- Get voltage prediction quickly
- Reduce testing time and cost

Efficient screening of potential electrode materials

- Quickly screen new materials
- Identify promising electrode materials
- Reduce experimental testing

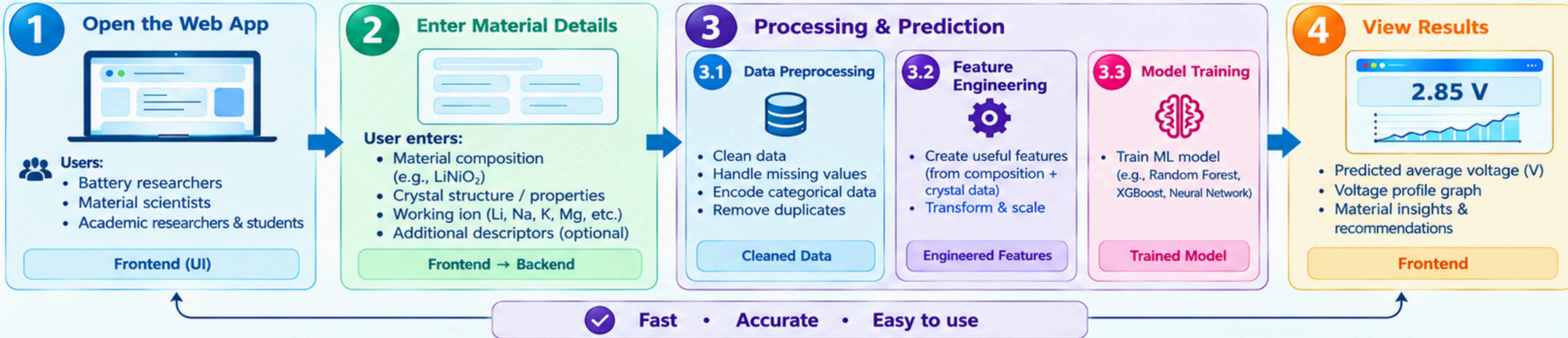


Machine Learning the Voltage of Electrode Materials in Metal-ion Batteries

Complete Project Architecture – From Data to Prediction



Smarter Materials → Better Batteries → A Greener Future



Example – What the User Sees

Input Form

Material Composition

LiNiO₂

Crystal Structure

Layered

Working Ion

Li

Additional Descriptor (optional)

e.g., Lattice constant, space group

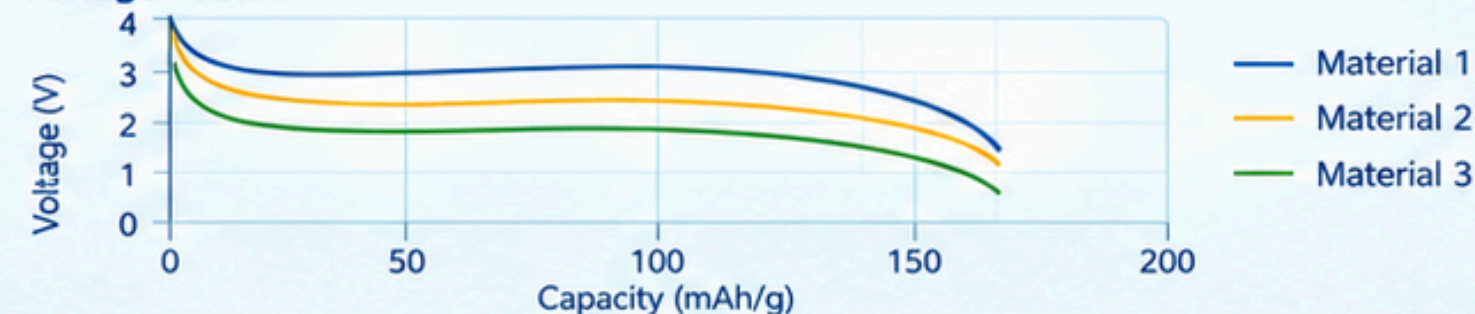
Predict Voltage

Result Page

Predicted Average Voltage

2.85 V

Voltage Profile



Additional Information

- Crystal structure: Layered
- Space group: R-3m
- Working ion: Li
- Predicted stability: High



This material shows good voltage performance and is a potential candidate for metal-ion batteries.



Models Used



Random Forest
(ensemble learning)



XGBoost
(gradient boosting)



Neural Network (MLP)
(deep learning)

Conclusion:

PROGRESS:

- Defined the ML problem and objective
- Analyzed the battery-voltage dataset
- Identified features and target
- Selected the ML models

NEXT MILESTONE:

“By the Next Milestone.We’ll Turn Our Concept into a Working Solution.”

“Compare different ML models based on their prediction accuracy using MAE, RMSE, and R^2 , and select the best-performing model.”

THANK YOU!

